1. Summary

This paper frames ML, AI and GenAI[1] in the energy sector not as a mere tool upgrade, but as a catalyst for a new operating model for data-driven critical infrastructure. It focuses on onshore/offshore wind power, associated grid operations, and Battery Energy Storage Systems (BESS) as the coupling element between generation volatility, grid constraints and market logic. The core proposition is as follows: the value of AI arises both from robust algorithmic methods and from the capability to turn data into operationally actionable evidence - under uncertainty, time pressure and complex constraints.

Many AI applications in the energy sector primarily reduce the cost of prediction: forecasts, failure probabilities and state indicators become faster and often more accurate. Value, however, emerges only when this prediction is translated into accountable decisions and actions—under grid-security, safety, liability and degradation constraints. Hence a useful distinction: prediction tasks deliver probabilistic information; decision tasks optimise objective functions under constraints. Between the two sit the real complementary investments: data integration, context modelling, MLOps/LLMOps, OT security, and the clarification of processes and roles.

The paper structures the discussion around three impact mechanisms: (i) *perception* (e.g. anomaly detection, pattern analysis, vision AI), (ii) *foresight* (probabilistic forecasts, remaining-useful-life and degradation models, simulation-based risk assessment), and (iii) *decision preparation* (robust optimisation, safe exploration, dispatch and resource scheduling). In parallel, it explains why "experimental data science" often fails precisely where production-grade AI must begin: in processes, roles, IT/OT architectures, MLOps/LLMOps, safety logic and auditability.

A particular emphasis therefore lies on an operational and governance perspective: objective systems in the energy sector are multi-dimensional; proxy KPIs can create perverse incentives; OT cybersecurity is a prerequisite for scaling; and GenAI requires provenance, observability and SLOs to become productive and responsible. On that basis, exemplar use cases are described - from yield and feed-in forecasting to predictive maintenance, wake steering and offshore O&M optimisation, through to grid-proximate event correlation, congestion/switching decision support, OT cybersecurity, GenAI for knowledge work, and BESS-specific control and value-stacking logics.

2. Context and strategic background

Generative AI in the energy sector is not merely a matter of new tools; it is a strategic infrastructure choice. Anyone deploying GenAI productively is implicitly deciding on dependencies along a chain comprising compute capacity (GPU/HPC), access to data, model providers, security and governance capability, and talent. In the European context, both data and scalable compute resources, reference applications, capital and legally robust trust environments may become bottlenecks. This leads to a two-track adoption logic. In the short term, one can harvest quick productivity gains in the "slipstream" of established models - especially in engineering knowledge work, documentation, analytical assistance and modelling workflows. In the medium to long term, however, it becomes decisive to build one's own capability: through sovereign data spaces, clearly defined operating environments, controlled model usage, and - where strategically justified - one's own models or at least one's own inference stacks. For critical infrastructure, time-to-value must not result in unreflective lock-in. A GenAI programme should therefore pursue reversibility by design from the outset: models, data and inference stacks should be designed so that providers remain substitutable, rather than treating this as an ex-post compliance exercise.

The deployment of machine learning, artificial intelligence and generative AI in wind energy and grid operations is therefore neither an end in itself nor a passing fashion. It is an expression of structural change in the energy system. As onshore and offshore wind expands, volatility in the generation portfolio rises, while requirements for availability, forecast quality, grid stability, operational control and regulatory evidencing increase. In addition, many assets—especially offshore - are difficult to access, capital-intensive, maintenance-heavy, and must be operated over decades. In such an environment, AI creates value not by replacing human expertise, but by combining data, models and decision logic so that states are detected earlier, risks are assessed more robustly, and actions are prepared more consistently.

AI's true relevance therefore sits at the intersection of uncertainty, time-criticality and complexity. Wind farms and grids are not static technical objects, but dynamic systems with stochastic influences, partially observable states, coupled constraints and often conflicting objectives. An offshore wind farm must simultaneously ensure high technical availability, economic yield, safe access, logistical controllability and grid-compliant feed-in. Grid operations, in parallel, must ensure security of supply, N-1 capability, voltage and frequency control, cyber resilience, regulatory compliance and economic efficiency. In both worlds, the critical question is not whether data exist, but whether data become decision-ready information.

Against this backdrop, three fundamental mechanisms can be distinguished. First, AI improves system perception, for example via anomaly detection, pattern analysis or image-based inspection. Second, it improves foresight, for instance via probabilistic yield and load forecasts, remaining-useful-life models or simulation-based risk assessment. Third, it

[1] GenAI (Generative Artificial Intelligence) refers to AI models - most commonly large language models - that generate new content from learned patterns, such as text, code, images, or structured responses, and are often "grounded" via retrieval (RAG) in vetted sources.

improves decision preparation, by deriving robust proposals for maintenance, dispatch, switching actions or operational scheduling from forecasts, constraints and objective systems. It is precisely at this third stage that experimental data science diverges from production-grade AI: in industrial operations it is not enough for a model to be statistically strong; it must be embedded in processes, roles, IT/OT architectures, safety rules, accountabilities and escalation chains.

Accordingly, the topic should not be understood as an isolated technology initiative, but as part of a new operating model for data-driven technical systems. The decisive governance question is: which objective is actually being optimised? Many AI projects fail not due to insufficient model performance, but due to an ill-defined objective function. If, for example, curtailment is reduced, O&M costs are lowered, or availability KPIs are improved, this can generate adverse incentives unless safety margins, lifetime behaviour, system strength, staff load or fairness aspects are also modelled. In the energy system, the gap between the true objective and the measurable proxy is therefore especially consequential. Good AI in wind and grid operations is therefore inherently multi-objective, constraint-aware and auditable by design.

3. Architecture and operating logic

A further core assumption of this compendium is that AI in industrial settings almost never works as a pure cloud solution. Real architectures are typically hybrid. Central platforms are suitable for fleet analytics, training pipelines, portfolio modelling, probabilistic forecasts and long-horizon optimisation. Edge-proximate systems are required wherever latency, data sovereignty, communication robustness or security zoning matter - e.g. in substations, converter stations, wind-turbine gateways or industrial network segments. Mobile devices play a role in service and maintenance contexts, particularly where diagnostics, checklists, image data or assistive functions are needed on site. TinyML becomes relevant where very small, low-power sensor nodes are intended to capture leading indicators before larger data streams are triggered.

Technically, the central design pattern is therefore often: local pre-processing and detection, central correlation and orchestration. A local system may detect an unusual vibration signature or a communication pattern without yet delivering the full causal diagnosis. The actual classification then occurs in a central platform that draws on historical data, fleet knowledge, environmental conditions, maintenance history and system topology. This pattern fits wind and grid operations well because it combines robustness, security zoning and scalability.

To make such a system viable in operations, MLOps is required as a genuine operating discipline. Models age; data regimes change; sensors drift; spare parts change subsystem behaviour; new operating modes emerge; and external drivers shift probabilities. In wind energy, weather regimes, seasonality, retrofits and OEM-specific differences are added; in grid operations, topology changes, protection concepts, load shifts, new feed-in sources and IT/OT interactions. A model that works well at commissioning may therefore become unreliable months later. Drift monitoring, revalidation, controlled rollouts, fallback mechanisms and documented intervention rights are not optional extras, but part of safety- and quality-compliant operations.

In wind farms, substations, converter stations and control rooms, AI cannot scale without OT cybersecurity. The reason is straightforward but consequential: in OT, availability and operational safety often matter more than “classic” IT security principles, and many IT-centric controls cannot be applied in OT without side effects. Once ML pipelines, edge analytics, remote access, cloud correlation or GenAI assistance enter technical operating processes, the attack surface expands - and with it the need for a systematic, sustainable OT security programme. This programme is less a one-off project than an operating discipline that reduces risk to an as low as reasonably practicable (ALARP) level and keeps it there. Practically, this requires (i) management commitment, (ii) a consistent framework integrating engineering, operations and security processes, and (iii) delivery discipline - covering monitoring, policy management, regular testing and clear responsibilities. In OT, without a robust security and governance foundation, AI remains pilotitis - or becomes an operational risk.

AI delivers value only when three conditions are met: is clear which decisions it supports, who owns those decisions, and how operations work with it. In critical infrastructure, decision rights should be made explicit: what remains human, what becomes AI-assisted, and what may be partially automated under guardrails. A practical structure is a three-step loop: (i) decision preparation (evidence, uncertainty, options), (ii) decision (accountable approval, intervention rights), (iii) execution and learning (feedback, post-mortem, model/process adjustment). This avoids both pilotitis and risky automation without a safety case.

4. Methodology: algorithmic classes, core definitions and data logic (wind & grid)

Anyone wishing to discuss AI/ML in wind energy and grid operations in an engineering-ready way should classify methods not primarily by "hype level" but by their mathematical structure. Figure 1 positions AI/ML/RL/DL at the intersection of statistics, IT, and operational practice (operations, automation, actuation).

In all cases, one seeks a mapping between observable variables x (features, measurements, context) and a target variable y (forecast, class, decision, cost function)—and in all cases the quality of input data determines whether the result is robust or merely plausible. In this sense, ML/AI is often "only" an extension of the classical modelling world: simple linear mappings are replaced by richer function classes; deterministic point estimates by probabilistic models; single-shot optimisation problems by learned or adaptive decision rules. The core remains: without clean measurement chains, consistent processes and a viable IT/OT architecture, even the best model is worthless.

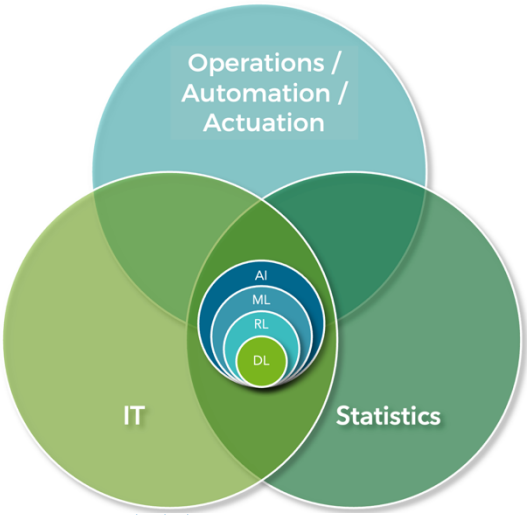

Figure 1: AI/ML/RL/DL at the intersection of statistics, IT and operational practice.

4.1 Closed-Loop Intelligence: Artificial Intelligence as a Discipline of Continuous Improvement in Critical Infrastructure

A mature artificial intelligence architecture in critical infrastructure is best understood as a closed-loop discipline rather than a sequence of isolated analytical tasks. Its value does not end with improved forecasts, sharper anomaly detection, or faster classification. Its real contribution begins where operational evidence is translated into structured managerial learning, where learning informs planning and process design, and where updated planning assumptions improve future operations. At this point, artificial intelligence moves from experimental capability into an operating discipline. This logic aligns closely with a continuous improvement programme, because it connects observation, interpretation, challenge, adaptation, and renewed execution in a disciplined cycle. In this sense, artificial intelligence supports a culture of intelligent improvement rather than a collection of technical point solutions.

This distinction matters because energy infrastructure operates under persistent uncertainty, long asset lifetimes, safety obligations, and coupled technical and commercial constraints. Wind farms, substations, converter stations, storage systems, and associated grid processes do not reveal their true operating character at commissioning. Their behaviour becomes visible over time through weather regimes, degradation pathways, maintenance interventions, congestion episodes, control actions, connection requests, and shifting market conditions. A technically serious artificial intelligence architecture therefore requires more than good model performance. It requires a structured learning loop that connects system perception, foresight, decision preparation, execution, validation, and renewed planning. That view is fully consistent with the wider argument of this compendium: prediction creates potential, while value emerges only when probabilistic information is converted into accountable action under real operational constraints.

Within such a loop, artificial intelligence supports four linked functions. First, it improves system perception by detecting patterns, anomalies, degradation signals, utilisation shifts, and weak indicators that would otherwise remain diffuse across operational data. Second, it strengthens foresight through probabilistic forecasts, state estimation, risk-oriented simulation, and forward-looking views of load, generation, failure likelihood, and operating margins. Third, it prepares decisions by evaluating feasible actions against technical, economic, contractual, and safety constraints. Fourth, and strategically most relevant, it helps institutionalise learning by showing which assumptions remain robust, which operating rules have become too conservative or too fragile, and where planning, maintenance concepts, or operating envelopes should be updated. This fourth function gives artificial intelligence its management relevance. It turns evidence into improvement.

Many organisations already generate the raw material for such learning, yet only a fraction is converted into structured intelligence. Repeated connection denials, recurring curtailment situations, seasonal capacity asymmetries, underused flexibility options, maintenance-driven bottlenecks, or persistent deviations between expected and actual behaviour often remain embedded in local workflows. Each item may appear operationally modest. Taken together over time, they form a high-value evidence base for better planning, more credible investment prioritisation, sharper operating rules, and more mature governance. A closed-loop intelligence model helps capture precisely this layer of value. It connects day-to-day operational traces with managerial judgement and long-horizon improvement.

For that reason, the central design question is less whether a model is mathematically sophisticated than whether the loop itself is well governed. A useful operating structure consists of five steps. First, evidence is captured from operations, engineering workflows, market interfaces, maintenance history, and context data. Second, artificial intelligence condenses this evidence into patterns, probabilities, and option spaces. Third, responsible managers assess the implications against objectives, constraints, and escalation logic. Fourth, decisions are implemented in planning

assumptions, maintenance concepts, operating envelopes, or operational procedures. Fifth, the consequences of these decisions are monitored and used to recalibrate both the model and the managerial rule set. This is the practical architecture of continuous improvement in a data-rich industrial environment.

Seen in this light, artificial intelligence becomes a force multiplier for disciplined management rather than a substitute for it. Human judgement retains a central role, especially where trade-offs must be made between availability, cost, degradation, safety margins, staff load, regulatory obligations, and resilience. In critical infrastructure, the decisive question is therefore whether a model output can be embedded into an accountable decision path. Decision rights, intervention authority, fallback procedures, and validation safeguards should be explicit from the outset. This compendium has already argued that productive artificial intelligence requires clear ownership, hybrid architectures, robust operations practices, and secure technical foundations. Closed-loop intelligence builds on that foundation and extends it into the domain of managerial learning.

The managerial benefit of this approach is considerable. It creates a more coherent relationship between operations and planning. It reduces the distance between evidence and consequence. It supports a culture in which assumptions are regularly challenged, patterns are interpreted rather than merely observed, and improvement becomes cumulative rather than episodic. For leadership teams, this matters because continuous improvement depends on more than reporting discipline. It depends on whether an organisation can turn experience into better future choices. Artificial intelligence can accelerate that transition, provided it is embedded in a structure that values traceability, reflection, and iterative adaptation.

In that sense, closed-loop intelligence is best understood as a practical architecture for continuous improvement in critical infrastructure. It gives technical systems a stronger memory, organisations a more disciplined learning rhythm, and management a more reliable basis for adaptation under uncertainty. Its strategic contribution lies in tightening the loop between operational evidence, managerial judgement, structured improvement, and updated planning assumptions. That is where artificial intelligence acquires lasting relevance: not as a one-off tool, but as a durable discipline of intelligent improvement.

Regression models: from linear baselines to probabilistic prediction

The entry point for many energy use cases is regression: a continuous value is to be predicted - power, feed-in, load, temperature, state indicators or costs. Linear regression remains both a didactic example and a practical engineering building block because it is transparent, stable and an indispensable baseline. Formally, parameters β (and optionally an intercept b) are chosen to minimise the sum of squared errors: $\min_{\beta,b} \sum_{i=1}^{n} \left(y_i - (\beta^\top x_i + b)\right)^2$.

From an input-data perspective, this is no different from multivariate regression: a feature vector x_i, a target y_i, and - crucially - a well-defined temporal mapping of which x belongs to which y. In wind energy this mapping is delicate because SCADA values, curtailment flags, outages, maintenance states and weather regimes are not cleanly separated in time. A model predicting power from wind speed may therefore learn not aerodynamics, but the frequency of curtailment or the signature of specific curtailment strategies. The seemingly trivial requirement - synchronised time axes, masked outages, known operating modes - is thus the real prerequisite for insight.

Where ML/AI goes beyond classical regression is less about the existence of inputs and more about (i) richer function classes and (ii) uncertainty. In operational settings, confidence bands are often more valuable than a mean. Probabilistic regression therefore models both the conditional expectation $E[y|x]$ and the distribution $p(y|x)$, or at least quantiles.

Practically, this immediately reveals whether the IT and process landscape is “AI-ready”: if forecast features come from NWP ensembles, SCADA, plant topology and grid states, it must be clear which weather-model version was used, whether curtailment was flagged correctly, whether measurements were revised ex post, and whether feature engineering is reproducible. Without data versioning, feature definitions and lineage, forecasts become arbitrary; with them, they become operational instruments.

Classification: events, fault patterns and decision triggers

Many of the highest-value applications are binary or categorical: failure yes/no, fault class A/B/C, anomaly vs normal operation. Logistic regression is the canonical starting point. It models a probability

$$p(y = 1|x) = \sigma(\beta^\top x + b), \quad \sigma(z) = \frac{1}{1+e^{-z}},$$

typically trained via cross-entropy / maximum likelihood. Here too, input data are not less critical - on the contrary, everything hinges on labels. In wind and grid operations, labels are rarely “laboratory labels”: they arise from ticket systems, maintenance reports, alarm texts, protection-relay logs, OEM codes and human judgement. If CMMS, SCADA, the control system and the asset register are not consistently integrated, no reliable y exists. Classification then becomes an attempt to derive “ground truth” from inconsistent text fragments - at high risk of spurious correlation. In practice, a useful engineering route is often to treat classification not as fully automated decision-making but as prioritisation: which events are unusual enough to escalate; which alarms are likely symptoms rather than causes; which combinations of state indicators and context signal incipient damage.

Trees, ensembles and robust non-linearity

Decision trees - and random forests in particular - are widely used in industrial environments because they handle mixed data types well and often deliver robust first results without excessive feature engineering. Mathematically, a decision tree approximates a function $f(x)$ by recursively partitioning the feature space; training typically aims to reduce impurity (for instance via entropy or the Gini index). A random forest then averages many such trees, commonly trained on bootstrap samples and with random feature sub-selection, which reduces variance and increases robustness:

$$\hat{f}(x) = \frac{1}{T}\sum_{t=1}^{T} f_t(x).$$

The practical drawback is rarely mathematical; it is organisational. As soon as several teams - or external suppliers - compute "similar but not identical" features, reproducibility collapses. At that point the model does not simply become worse; it becomes difficult to interpret and to trust. One can no longer tell whether performance improved because the model improved or because the data pipeline changed silently. This is precisely where industrial ML diverges from classical statistics in everyday practice: without disciplined feature definitions, versioning, schema management and monitoring, production quality becomes unknowable even when the algorithm is sound.

Unsupervised Learning: regimes, states and anomalies without labels

A large share of operational reality is not cleanly labelled. One may sense that something is "different" without being able to name it reliably. This is where unsupervised methods such as clustering, dimensionality reduction and autoencoders become practically useful. In k-means clustering, the goal is to partition data into k clusters by minimising the within-cluster sum of squared distances to the cluster centres:

$$\min_{\{\mu_j\}} \sum_{i=1}^{n} \min_{1\le j\le k} |x_i - \mu_j|^2.$$

In wind-farm operations, clustering can help separate regimes such as normal operation, curtailment patterns, icing-like signatures or sensor-drift clusters. In grid operations, it can help identify recurring patterns in event and measurement data associated with particular topologies or load situations.

Principal Component Analysis (PCA) compresses correlated variables into a smaller set of principal components that explain most variance. Formally, this can be derived via the eigenvectors of the covariance matrix Σ:

$\Sigma w = \lambda w$, where the leading eigenvectors w define the principal directions.

Autoencoders go one step further by learning a compressed latent representation $z = g_\phi(x)$ and reconstructing $\hat{x} = h_\theta(z)$, typically by minimising a reconstruction loss such as:

$$\min_{\phi,\theta} \sum_i |x_i - h_\theta\left(g_\phi(x_i)\right)|^2.$$

Anomalies then manifest as unusually high reconstruction error or atypical locations in latent space. The decisive issue, however, is not the mathematics but the definition of "normal". If the "normal" dataset already contains hidden faults, maintenance artefacts or systematic curtailment behaviours, the model will learn precisely that world as normal. Unsupervised learning therefore benefits disproportionately from clearly defined operating modes and governed data quality (completeness, plausibility, freshness and consistency).

Bayesian methods and uncertainty: from point estimates to distributions

In wind and grid operations, uncertainty is not an exception; it is the baseline condition. Bayesian methods make this explicit by treating parameters θ probabilistically and updating them after observing data D via Bayes' rules:

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}.$$

Their practical value is less philosophical than operational: they help distinguish what the model genuinely knows from where it extrapolates. This matters particularly for rare events - severe disturbances, unusual failure modes, extreme weather - where deterministic models may appear confident while in fact operating outside their empirical support.

The operational vulnerability is equally concrete. If measurement definitions, sensors or calibrations change, the error model changes; the likelihood no longer fits; and the posterior becomes misleading. Without data lineage and change logs, such breaks tend to be detected late - often only after decisions have already deteriorated. Bayesian approaches therefore reward disciplined system governance and punish silent system drift.

Gaussian processes, Bayesian optimisation and safe exploration

At this point the bridge to classical optimisation becomes very direct. Many operational optimisation problems in wind and grid contexts are (a) too expensive to evaluate freely, (b) too uncertain to trust naïvely, or (c) too risky to explore without constraints. Surrogate modelling and Bayesian optimisation address precisely this: instead of "solving" the original problem at full fidelity, one learns a probabilistic surrogate function that provides both a mean estimate and an uncertainty estimate and then selects evaluation points to learn efficiently.

Gaussian processes (GPs) formalise this as a distribution over functions: $f(x) \sim \mathcal{GP}\big(m(x), k(x,x')\big)$.

After observing data, the GP posterior yields a predictive mean $\hat{\mu}(x)$ and variance $\widehat{\sigma^2}(x)$, enabling an explicit exploration-exploitation trade-off. In safety- and liability-relevant environments, exploration cannot be unconstrained. "Safe optimisation" introduces confidence bounds $l(x)$ and $u(x)$ and enforces safety constraints so that physical limits - loads, voltages, temperatures, operating boundaries - are respected with high probability.

Wake steering or controller tuning illustrates the logic well. One may aim to increase AEP, but must not violate load limits, stability margins or operational guardrails. This requires context-rich data (wind direction, turbulence, operating mode, curtailment and topology) and disciplined change management; otherwise effects cannot be attributed causally and optimisation degenerates into guesswork.

Reinforcement Learning: sequential decisions – only with a safety case

Reinforcement learning is the most strongly "AI-labelled" class of methods, and it comes with the strictest requirements in critical infrastructure. RL typically frames problems as a Markov decision process and seeks a policy that maximises expected discounted return. In compact form, the objective is: $\max_{\pi} \quad E[\sum_{t=0}^{\infty} \gamma^t r_t]$, with discount factor $\gamma \in (0,1)$.

A canonical update rule in value-based RL is *Q-learning*, which updates an action-value estimate according to the Bellman target: $Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a'} \quad Q(s',a') - Q(s,a)\right]$.

This is appealing because many operational problems are inherently sequential: dispatch, campaign scheduling, wake steering, or congestion management over time. The engineering constraint is simple: classical RL learns through exploration, and exploration implies the possibility of unsafe actions. In critical infrastructure, this is acceptable only under stringent conditions - typically simulation-first approaches supported by a digital twin, conservative guardrails, stress testing and controlled approval processes. In practice, RL is therefore most credible as a simulation-trained advisory component (or, in rare cases, as tightly bounded controlled automation), not as a free-learning operational controller. Without a twin, data consistency and change discipline, RL is not merely difficult; it is irresponsible.

GenAI and RAG: provenance-grounded knowledge work

Generative AI is particularly valuable in wind and grid contexts when it structures knowledge work: manuals, standards, as-built documentation, maintenance reports, lessons learned, alarm texts, specifications, and minutes. The central pattern for productive use is retrieval-augmented generation (RAG): rather than letting the model hallucinate freely, context is retrieved from a controlled knowledge base. Technically, documents are mapped into embeddings $e(d) \in R^k$ and a query q likewise. The system retrieves the top-K documents by similarity, for example via cosine similarity $\mathrm{sim}(q,d) = \frac{e(q) \cdot e(d)}{|e(q)||e(d)|}$, and generates the answer conditioned on these sources.

In industrial operations, the critical question is not "can the model answer?", but "is it permitted to - and is the answer provable?". Provenance, permissions, audit logs, role models, data classification and currency are the core conditions. If the document landscape is fragmented, if there is no single source of truth, if versions are unclear, or if access rights are not modelled cleanly, GenAI will not scale - or it will scale risk. RAG is therefore less an NLP trick than an architectural principle: controlled information delivery with source binding.

5. The common denominator: data are both input and part of the system

If one examines these algorithmic classes soberly, the observation is confirmed: input data are existential in much the same way as in multivariate regression or MICQP - except that requirements in ML/AI are often stricter, because models are adaptive, deployed at scale, and live in non-stationary environments. In wind and grid contexts, data are part of a socio-technical system: both raw material and an operating foundation. Without clear processes, accountabilities, and an IT/OT integration architecture, three typical pathologies emerge.

- First, the *pathology of temporal inconsistency*: sensors, SCADA, CMMS, weather models and grid events operate on different time scales and time stamps. Without defined synchronisation logic, any causal interpretation becomes fragile.
- Second, the *pathology of missing context*: operating modes, maintenance states, firmware versions, curtailment and topology changes are not cleanly maintained as context variables; models then learn "operating policy" rather than "system behaviour".
- Third, the *pathology of non-reproducible features*: if feature generation is not versioned, audited and monitored, models become artefacts of individual notebook runs. This is where the operational truth of "AI readiness" lies: not in the choice of algorithm, but in the ability to run data, features, models and decisions as an end-to-end chain.

One might put it this way: ML/AI extends the classical modelling world with richer function classes and explicit uncertainty logic - but governance and data requirements do not remain constant; they increase. Or, in other words, ML/AI is rarely "new mathematics", but almost always a new operating discipline for mathematics in living systems.

Many use cases become operationally viable only through a digital-twin logic. A digital twin is less a 3D model than an operationally usable representation of an asset, a process or a grid state, used to simulate behaviour and validate decisions. Three functions are central: (i) a state/belief twin (e.g. SOC/SOH, degradation), (ii) a physical/control twin

(wake, load flow, inverter-based dynamics), and (iii) an operational twin (O&M planning, resources, approvals). The twin enables shadow tests, stress tests and what-if analyses - and thereby a safety case and evidencing.

Responsible AI should not be treated in the energy sector as an abstract ethics debate, but as an ongoing governance function that connects technical, regulatory and operational requirements. In critical infrastructure, this concerns concrete properties: traceability, contestability, robust validation, abuse prevention, data protection, and clear accountability across the full lifecycle of data, models and decisions. Precisely because many energy data are person- or operations-adjacent (e.g. grid-proximate measurements, customer load profiles, operational incidents), and because mis-decisions can have real physical consequences, Responsible AI must be anchored as a business-as-usual discipline - not as a one-off policy. In this logic, edge AI gains additional significance: models operated close to the asset or within security zones can meet privacy, latency and resilience requirements more effectively, but are responsible only when embedded in a consistent governance and monitoring regime. The transition from pilots to production systems therefore depends not merely on model quality, but on whether organisations can integrate accountability, evidence, risk control and operational safety.

6. Concrete wind and grid application examples

The adjacent figure serves as a map of the use cases described in this paper. It organises them along key capability classes of AI/ML/GenAI - from sensing and forecasting, through state estimation, to optimisation, decision support, and knowledge-based assistance.

The figure does not claim to be exhaustive; rather, it serves as a structural aid: it highlights which methodological building blocks dominate in which use cases, and where shared data and operational requirements arise. On this basis, the following use cases are discussed in detail.

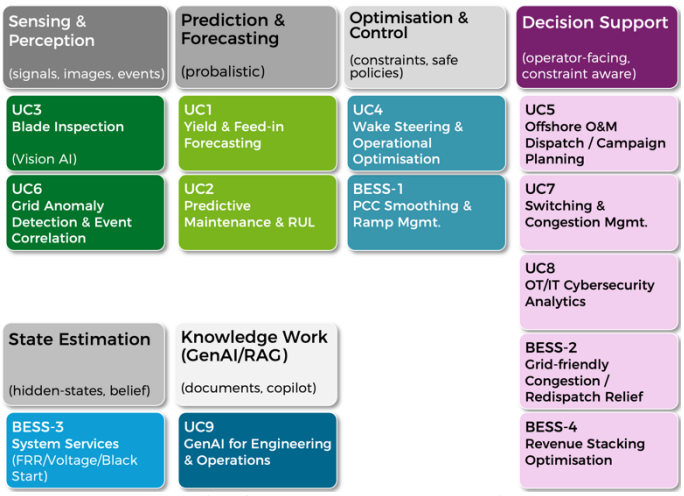

Figure 2: Map of wind/grid/BESS use cases organised by AI/ML capability class

Use case 1: yield and feed-in forecasting

Yield and feed-in forecasting is one of the most obvious and, at the same time, one of the most technically demanding ML applications in the wind domain. Whereas classical approaches often rely on deterministic weather assumptions and simplified power curves, modern ML methods allow a more differentiated linking of weather ensembles, turbine states, curtailment situations, wake effects and historical operating regimes. The decisive point is less the point forecast itself than the ability to model uncertainty explicitly. For operational decisions, it matters both how much power is likely to be generated tomorrow and with what probability certain deviations may occur. Only on that basis can balancing-energy needs, redispatch risks, or more conservative operating strategies be managed more effectively.

Technically, this implies that probabilistic models, quantile estimators, calibrated sequence models, or ensemble-based approaches are often better suited than pure mean predictors. This becomes especially interesting across multiple levels at once: turbine, plant, cluster, portfolio and grid node do not follow identical dynamics. A sound model architecture must therefore capture local effects - such as wake interactions and turbine behaviour - as well as systemic influences from weather, curtailment or grid congestion. In practice, the economic impact of such models can be substantial because forecasting errors in power systems generate asymmetric costs. A small improvement in forecast quality can therefore be economically far more valuable than a large improvement in a purely statistical metric.

Use case 2: predictive maintenance and remaining useful life

Predictive maintenance is particularly valuable in the wind domain because unplanned outages generate both repair costs and, frequently, substantial opportunity costs through downtime, secondary damage and complex offshore interventions. The challenge is that many critical degradation processes are only indirectly observable. Bearings, pitch systems, converters, generators or drivetrain components often exhibit early anomalies only in weak signals, noisy spectra, or multivariate correlations. ML becomes effective here when it does not merely classify “faults” but identifies degradation pathways and derives a robust maintenance logic from them.

Methodologically, the spectrum ranges from classical anomaly detection to survival-based approaches and remaining-useful-life (RUL) models. A key practical issue is that many real-world datasets contain only a small number of clean failure labels. Practice is characterised by incomplete maintenance reports, changing component configurations, sensor gaps and confounders. A bearing may be replaced before it fails; a converter may behave differently after a firmware change; maintenance activity may influence the signal signature more strongly than

ageing itself. Engineering-ready AI must therefore cope with incomplete, noisy and partly biased labels. It must not only detect patterns but distinguish useful early-warning signals from spurious correlations.

This use case is especially valuable offshore because the operative decision is not “fault yes or no”, but rather: when do we intervene, with which team, with which material, in which weather window, and within which campaign logic? Predictive maintenance therefore delivers its greatest impact not as an isolated diagnostic model, but as an input into higher-level dispatch and logistics optimisation.

Use case 3: rotor-blade inspection and vision AI

Rotor blades are highly stressed components whose condition directly affects yield, load behaviour, lifetime and safety. Classical inspections are time-consuming, labour-intensive, and their quality often depends heavily on experience, weather, visibility and individual judgement. Vision-AI-based methods promise not simply automation, but a standardisation of perception. Image- and thermography-based approaches can identify leading indicators of erosion, cracks, delamination or lightning damage with high consistency, thereby focusing attention on critical findings.

The core technical challenge, however, lies not only in image classification, but in the variability of the data-acquisition process. Perspective, lighting, soiling, surface condition, camera characteristics and blade geometry can all influence the outcome substantially. Good vision AI must therefore be robust under conditions that are difficult to reproduce cleanly in a laboratory. At the same time, a high average detection rate is insufficient if safety-critical defects are missed. The operational quality of such a system is therefore measured less by an abstract model metric alone than by whether relevant findings are escalated reliably, documented cleanly, and translated into robust maintenance decisions. For precisely this reason, humans remain accountable for critical classes - not as a sign of technical weakness, but as an expression of a sound safety case.

Use case 4: wake steering and operational optimisation

Optimising wind-farm operation is among the most demanding AI applications in wind energy, because aerodynamics, structural loading, turbulence, grid constraints and economic objectives interact directly. Wake steering is a typical example: a turbine is deliberately yawed or operated in a way that modifies its wake to improve conditions for downstream turbines. This can increase overall park yield, yet it simultaneously changes local loads, fatigue effects and uncertainty. The problem is therefore not purely data-driven, but a hybrid optimisation problem at the intersection of physical modelling, statistical uncertainty and operational constraints.

This is where it becomes particularly clear that AI in an engineering context must not be blindly exploratory. Reinforcement learning may sound attractive, but in real plants it is meaningful only under stringent conditions. Learning by trial and error is acceptable in safety- and liability-relevant systems only where simulation, a digital twin, conservative guardrails and controlled approval processes are in place. In practice, more robust approaches are often surrogate-based optimisation, model predictive control, or Bayesian methods that treat uncertainty explicitly and avoid unsafe exploration. The realistic maturity path therefore almost always runs through simulation and shadow mode before operational recommendations - or automated set-points - are even considered.

Use case 5: offshore O&M, dispatch and campaign planning

Offshore wind is both an asset and a logistics system. Weather windows, vessel capacity, crew availability, access to spare parts, safety approvals and technical priorities form a tightly coupled operating problem. Classical planning approaches often respond locally and event by event, whereas the real lever lies in proactive campaign formation. AI and advanced optimisation create value here by translating technical findings, weather forecasts and resource constraints into rolling decision models.

It is particularly important that such models must not simply minimise cost. A system that shifts missions aggressively may appear economically attractive in the short term, yet it can generate safety risks, follow-on failures or systematic overload of crews. This use case therefore illustrates why multi-objective optimisation is so central in energy systems. Availability, safety, cost, plan stability and spare-parts logistics are not alternative objectives; they must be considered jointly. Good models do not create a false sense of precision; rather, they help set better priorities under time pressure and tight constraints.

Use case 6: grid-proximate anomaly detection and event correlation

In grid operations, data-driven methods add particular value where many inconspicuous signals combine into a meaningful pattern. Individual protection-relay events, communication disturbances, time-synchronisation issues, PMU deviations or unusual load-flow patterns are often harmless when viewed in isolation. In combination, however, they may indicate incipient faults, measurement errors, misconfigurations or cyber indicators. AI’s added value lies in correlating heterogeneous events across time, topology and system boundaries.

Methodologically, this is a domain in which time-series analysis, graph models/graph neural networks (GNNs), and rule-based domain logic must be brought together. Pure black-box models are rarely sufficient, because in an incident explainability, escalation logic and trustworthiness are decisive. The best solution is therefore often not fully autonomous classification, but an advisory layer that prioritises conspicuous patterns, condenses them visually, and enables operators to act faster and more robustly. In grid operations, this is frequently more effective than attempting to remove humans from the loop.

If AI in grid operations is taken seriously, it should be framed not merely as “alarm correlation” or “decision support”, but as an enabler for grid-enhancing technologies that combine physical flexibility with digital controllability. These include dynamic line rating, intelligent curtailment (instead of blanket curtailment), optimised reactive-power provision and voltage control, and increasingly adaptive operating strategies based on probabilistic forecasts - including extreme-value considerations for wind and load. The underlying logic is systemic: the more volatile and decentralised the system becomes, the more the bottleneck shifts from “hardware alone” to “hardware plus digital operations”. This is not, however, plug-and-play. The real hurdle often lies in interoperability, communications infrastructure, data access and an end-to-end platform logic that brings together measurements, topology, protection/control events and operating constraints. Only then can data-driven methods deliver their impact - for example by identifying safe operating headroom that would otherwise remain unused out of caution, or by progressing stepwise towards autonomous reconfiguration and collaborative asset management.

Use case 7: switching actions and congestion management

Supporting switching actions and congestion management is among the most sensitive AI use cases of all. Here, physical constraints, regulatory requirements, safety logic and liability considerations collide directly. The value of AI does not lie in “automatic grid control” in any simplistic sense, but in the capability to derive robust courses of action more quickly from historical situations, current state data, load-flow models and uncertainty information.

A technically sound approach couples forecasting, state estimation and optimisation. The model must therefore both predict what is likely to happen and evaluate which option is robust under the prevailing constraints. In partially observable situations - for example with incomplete sensor coverage, uncertain feed-in trajectories or unclear equipment availability - belief-based decision-making is particularly helpful: the system represents uncertainty about grid state explicitly and chooses actions that are robust to that uncertainty. The operational threshold for automation nevertheless remains high. Precisely because these use cases cut deeply into operational responsibility, traceability, contestability and clear guardrails matter more than maximum algorithmic “elegance”.

Use case 8: cybersecurity in converged OT/IT environments

As wind farms, substations and converter-adjacent infrastructure are digitised, an increasingly dense mesh emerges of remote access, APIs, data pipelines, central analytics platforms and distributed edge components. This connectivity provides the breeding ground for productive AI - while simultaneously creating new attack surfaces. AI in the cyber context is therefore relevant in two ways: first as a tool for detecting unusual communication and access patterns, and second as a design problem for secure architectures. The benefit is not merely to detect attacks, but to contextualise anomalies early enough that operations, security and asset responsibility can act jointly and effectively.

In OT environments in particular, security must not be pursued at the expense of availability. Overly aggressive blocking can itself become an operational risk. The right balance is therefore required here as well: segmented architectures, well-structured identity and access management, auditable data flows, and clearly defined response pathways. In this domain, AI is not a substitute for sound security architecture; rather, it is an amplifier where pattern recognition extends beyond the limits of human perception.

Use case 9: GenAI for engineering and operations

Generative AI has a different character in wind and grid contexts from classical ML use cases. The focus is usually not real-time decisions within the technical process, but the productivity of knowledge-intensive work. Engineering documents, operating manuals, maintenance reports, lessons learned, alarm texts, specifications and approval documents form a knowledge space that is difficult to access in many organisations. GenAI can unlock this space by making information easier to find, summarise and translate into action contexts. The core value lies in accelerating qualified knowledge work, not in unconstrained text generation.

For GenAI to be viable in this environment, however, the provenance of answers must be secured. In industrial operations, a linguistically plausible response is not sufficient; every recommendation, summary or analysis must be traceable back to reliable sources. Retrieval-augmented generation, role-based access, confidential operating environments and logging are therefore central prerequisites. GenAI is particularly strong where it helps people with standardisation, pre-analysis, documentation and knowledge access. It is weak where source quality, liability, or technical consequence make an unverified hallucination unacceptable.

A particularly practical GenAI application in the energy sector is not free text generation, but an interaction layer to models: GenAI as an "energy-modelling copilot". In planning and transformation programmes, teams work with complex simulation and optimisation environments - often for months - under incomplete data and with many stakeholders. A conversational interface that supports model configuration, data understanding, scenario design and result interpretation can raise productivity substantially, not least because it opens access to modelling work for a broader range of qualified users. The boundary must, however, be explicit: such copilots are primarily intended for planning, scenarios and what-if analyses, not as real-time autopilots for operations. Especially under "imperfect data", GenAI can help to make gaps visible, document assumptions transparently and structure plausible options - provided provenance, uncertainty labelling and validation logic are implemented consistently. Technically, this forms an ideal bridge between digital twins, optimisation and knowledge work: the copilot connects data spaces, model libraries and documentation without itself becoming the decision-maker.

Operationally, LLM-based systems differ from classical software components: they are non-deterministic, their inputs are effectively unbounded in variability, and failure rarely appears as a clean "crash", but rather as quality drift, unexpected hallucinations, parsing errors or gradually increasing latency. Observability for LLM systems is therefore not a luxury, but the prerequisite for stable use. A production-grade approach begins with telemetry over inputs, outputs, error classes, parsing results, tool calls and user feedback, and then turns these signals into an operating logic with measurable service-level objectives (SLOs). In many cases, two SLO classes are a pragmatic starting point: (i) latency (p95/p99) and (ii) error rates or task success rates, complemented by quality proxies (e.g. citation rate to sources, share of refused answers where evidence is missing). This is particularly important for agentic workflows in which models execute chains of actions: without constraints, "unbounded execution" can arise, causing costs, time and risk to increase uncontrollably.

7. Battery Energy Storage Systems (BESS) as a bridge between wind power, grid operations and markets

BESS extends the wind-and-grid perspective by adding an element with a distinctive property in the energy system: it can provide power bidirectionally (charging and discharging), it can do so very quickly, and - depending on design and control - it can fulfil both energy-economic and grid-technical functions. This makes BESS the "coupling element" between three logics that are often treated separately in operations: (i) generation volatility (wind), (ii) grid-operational constraints (congestion, voltage, system strength), and (iii) market logic (arbitrage, balancing services, capacity, contracts). It is precisely this coupling that makes BESS AI- and ML-intensive - not because the mathematics is new, but because several objective systems overlap simultaneously and dynamically.

8. BESS use cases: why AI/ML is particularly "natural" here

(a) Renewables integration, smoothing and firming ("smoothing wind" at the grid connection point)

Operationally, BESS means the following for wind farms: fluctuations at the point of common coupling (PCC) are smoothed, ramps are reduced, forecast errors are buffered, and - under certain conditions - feed-in can be shifted in time ("firming"). This function gains systemic importance because future grids will be more decentralised, more volatile and more dynamic. Hybrid energy storage (HESS) explicitly addresses the fact that individual storage technologies are not optimal across all time scales and discusses sizing and power-management strategies to maximise the smoothing effect.

AI/ML comes into play where the control problem is not deterministic: wind and load regimes change, forecasts carry uncertainty, the storage system faces SOC and power limits, and the optimal operating strategy depends on a target mix of smoothing, cost, degradation and market revenues.

(b) System services: frequency, voltage and inertial services through to black start

In grid operations, BESS is becoming increasingly relevant as a provider of system services. The short study on the system value of large batteries explicitly mentions instantaneous reserve (as an immediate response to power imbalances) - historically provided by rotating masses and increasingly by converter-based assets in the future - and notes that batteries can in principle provide this but then require grid-forming inverters. It also describes black-start capability as a role for large batteries, particularly relevant in regions without suitable conventional power plants.

From an ML/AI perspective, these are not "nice-to-have" add-ons, but highly dynamic control and state problems. Frequency and voltage support must operate under measurement noise, communication latency, modelled uncertainty and strict safety limits; black-start logic requires robust sequences, approvals and evidencing.

(c) Congestion and redispatch impact ("grid-friendliness" rather than market-friendliness alone)

The study on system value makes the point clearly that batteries in Germany are currently operated predominantly on a market basis, and that the potential to reduce grid costs - especially through congestion reduction and hence redispatch - is not exploited systematically, among other reasons due to missing incentives within a uniform price zone. It also identifies external effects: impacts on redispatch, the congestion implications of short-term set-point

changes, perverse incentives within the 15-minute settlement interval, schedule jumps between MTUs, and effects on voltage control.

This describes the AI/optimisation problem almost perfectly: a multi-objective, constraint-heavy, time-discrete decision problem in which a purely market-driven objective (arbitrage/balancing services) can lead to systemically unfavourable grid outcomes. The study also points to classes of instruments that address precisely this dimension (regionalised signals, grid-connection guardrails, ramp limits, earlier gate closure, and so forth).

(d) Revenue/value stacking as the operational reality

Once BESS is in the system, it is typically not run economically through a single use case, but through value stacking: arbitrage plus frequency services plus congestion/grid services plus, where applicable, capacity and contractual elements. Schmidt & Staffell describe "revenue stacking" as a structural value principle and provide a systematic map of applications (including frequency response/regulation, voltage support, black start, congestion relief, T&D deferral, renewables firming/smoothing, and arbitrage).

Alexa Capital argues similarly from an investor perspective: current returns often rest on a diversified value stack (arbitrage plus grid/utility services and, where applicable, capacity payments), whereas pure arbitrage alone is frequently not stable enough in many markets.

Technically, this means that the operating strategy must be not only physically feasible, but also compatible with market rules, grid constraints, and warranty/degradation requirements - an ideal environment for data-driven dispatch and policy optimisation.

9. Where exactly do the AI/ML algorithms sit in the BESS stack?

BESS is particularly interesting because it concentrates the classic triad - estimate, forecast, decide - in a compact form.

(i) State estimation: SOC/SOH as a hidden-state problem (partially observable)

The most important operational state of a battery system - *state of charge* (SOC) and *state of health* (SOH) - is not directly measurable; it must be estimated from currents, voltages, temperatures, impedance/OCV characteristics, cell-balancing information and operating history. Mathematically, this is a typical hidden-state problem: a (non-linear) state x_t (SOC, internal parameters) evolves according to a dynamics model, and measurements y_t (V, I, T, and possibly impedance proxies) provide only indirect information. In practice, filter families (Kalman variants/particle filters) are combined increasingly with ML-based surrogates; the key point is that model and measurement errors are not constant, but depend on temperature, C-rate, ageing and operating mode.

What data need to be available? High-resolution time series of I/V/T (ideally at string/cell level, or at least rack level), operating modes, BMS logs (balancing, limits, alarms), ambient/container climate, and event data (trips, limit hits, firmware changes). Without this granularity, SOC/SOH remains an estimate "in the fog", and downstream ML/optimisation decisions become unstable by necessity.

Typical IT/process difficulty: in many projects, BMS data and EMS/SCADA data are not fully integrated (time-stamp drift, different sampling rates, missing historisation). The result is that reports can be produced retrospectively, but reliable real-time control is difficult - and degradation cannot be attributed cleanly (was it temperature, C-rate, a specific market dispatch, or a BMS limit?).

(ii) Degradation and lifetime models: RUL as a techno-economic constraint

The second algorithmic core layer is degradation. The optimal operating strategy of a BESS is not only a function of prices and grid conditions, but also a function of lifetime cost and warranty constraints. Operationally, every operating strategy is a trade-off between short-term benefit (revenues/system services) and long-term asset consumption (capacity fade, resistance increase, higher failure probability, warranty risk). Structurally, this mirrors predictive maintenance in wind - except that in BESS, degradation is more strongly driven by operating policy and is therefore directly optimisable.

What data need to be available? Cycle counting (equivalent full cycles), depth of discharge, C-rate profiles, temperature profiles, time spent at high SOC, events (over-temperature, over-voltage), maintenance/partial replacements. In addition, one needs a stable mapping between "duty profile" and "state-related outcome" (SOH, capacity, internal resistance).

Typical IT/process difficulty: if market/grid dispatch and BMS data run in separate systems, a governance gap emerges. The organisation may optimise, for example, for revenues and only later discover that it has effectively "optimised against" degradation and warranty constraints. This is the proxy trap in its purest form—except that here it becomes directly monetary.

(iii) Dispatch and optimisation: multi-objective, stochastic, constraint-heavy

The third layer is the decision itself: when to charge, when to discharge, when to hold reserve, what ramps to run, what SOC targets to set, and how to respond to grid constraints. Conceptually, this is very close to an MIQP/MICQP

worldview: constraints (SOC dynamics, power limits, efficiency, ramps, grid limits, market products) plus an objective function (revenue stack, grid-friendliness, degradation cost, risk). The difference is not the mathematics per se, but the degree of uncertainty and non-stationary boundary conditions: prices and congestion states are stochastic, forecasts are error-prone, and grid constraints can arise at short notice.

In this layer, ML/AI is particularly effective as (a) a probabilistic forecaster (prices, congestion indicators, wind/load), (b) a surrogate for fast what-if evaluation, (c) policy learning (only with guardrails), and (d) anomaly or constraint detection to identify risky operating strategies early.

10. Concrete BESS application examples

BESS use case 1: wind PCC smoothing and ramp management (with a HESS option)

In wind farms or wind-cluster settings, a BESS can be controlled deliberately so that feed-in at the PCC becomes "calmer": short-term fluctuations (seconds/minutes) are damped, ramps are limited, and - where appropriate - energy is shifted to smooth forecast feed-in profiles. The HESS approach is particularly interesting here because it explicitly addresses multi-timescale capability: different storage technologies complement one another in covering both fast and slower variability, and the power-management strategy becomes the central optimisation problem.

Data: SCADA/wind/NWP ensembles, PCC measurements (P/Q), BESS I/V/T/SOC, grid constraints (ramp limits, reactive-power requirements).

Difficulty under weak IT/process conditions: if PCC measurements and BESS control are not coupled in real time (latency, time stamps), a control problem emerges: the BESS reacts too late or to the wrong signals, worsening stability rather than improving it.

BESS use case 2: grid-friendly congestion management / redispatch relief (co-optimised)

The system-value study makes the point clear: the largest grid-side lever potentially lies in reducing congestion (redispatch costs, grid reinforcement), yet this is not realised systematically today. An engineering-ready use case would therefore be a co-optimised operating strategy that integrates market and grid signals: the BESS charges not only when price is low, but also when it relieves the grid (or does not discharge when it would exacerbate congestion). At the same time, short-term set-point changes must be controlled, because they can themselves trigger congestion.

Data: congestion indicators (e.g. PTDF-based sensitivities, TSO/DSO redispatch signals), grid state/topology, market price curves, BESS operating data, gate-closure/MTU logic.

Difficulty: without clear rules and instruments (regionalised signals, guardrails, ramp limits), the problem remains structurally unresolved; the study lists precisely such instrument families.

BESS use case 3: system services—instantaneous reserve, frequency support, voltage control, black start

As rotating mass declines, inverter-based contributions become more important. The system-value study explicitly cites instantaneous reserve from 2026 onwards with remuneration by the German TSOs (~€50–70/MW·h, first tenders in Q1/2026) and the need for grid-forming inverters, as well as black-start potential. This means that AI/ML is not only about forecasting, but also about control, safety case and monitoring.

Data: high-frequency measurements (frequency, df/dt, voltages, phase angles, P/Q), BESS inverter data, protection/control events, synchronisation quality.

Difficulty: in many organisations, protection and control data, the BESS EMS, and market/dispatch systems are organisationally separated. This creates "breaks in responsibility" and makes auditability and incident analysis more difficult.

BESS use case 4: revenue stacking as an optimisation problem (market + technical + regulatory)

Schmidt & Staffell provide a map of applications and the principle that storage maximises value by serving multiple services simultaneously ("revenue stacking"). Alexa Capital similarly argues from an investor perspective that large batteries are often attractive precisely because arbitrage revenues can be combined with additional services - explicitly including frequency regulation, congestion relief and other ancillary services as part of the stack.

Engineering-ready in this context means: operators need an operating strategy that manages SOC availability for multiple markets at once (e.g. holding reserve versus running arbitrage) while pricing in degradation and warranty constraints.

Data: price time series (DA/ID), product definitions (balancing services), activations, grid signals, BESS constraints, warranty/degradation cost models.

Difficulty: if the IT landscape lacks a consistent "single schedule source" and does not provide transparency on SOC and constraints, internal conflicts arise (trading vs operations vs asset health) that can undermine almost any ML-supported optimisation in practice.

11. Closing remark

The introduction of AI in wind energy and grid operations should be understood neither as a short-term IT project nor as a visionary future narrative. It is part of a deeper transformation of technical operations: away from reactive, fragmented and implicit decision logic, and towards a more explicit, data-driven, probabilistic and auditable mode of action. An organisation's true maturity is therefore not measured by how many models it develops, but by whether it can translate models, processes, people, safety logic and governance into a viable operational architecture.

Annex I - definitions

1) Linear regression (forecasting / regressions baseline)

Mathematical definition

Given training data (x_i, y_i) with $x_i \in R^d, y_i \in R$. Linear regression estimates parameters $\beta \in R^d$ (and optionally an intercept b) by minimising the squared error: $\min_{\beta,b} \sum_{i=1}^{n} \left(y_i - (\beta^\top x_i + b)\right)^2$.

Practical example (wind/grid)

- *Wind*: Kurzfristiger Zusammenhang zwischen Windgeschwindigkeit (oder NWP-Features) und Leistung – als Baseline oder als Feature in komplexeren Modellen
- *Grid*: relationship between load $L(t)$ and price / congestion indicators (e.g. regressions illustrating price vs. load)

Data required

Time-synchronised feature vectors (SCADA / NWP / load-flow) plus a target variable (power, load, congestion cost, etc.). *Crucial*: consistent time slices and clean flags (curtailment, outages), otherwise the model "learns" disturbances as physics.

Typical pitfalls under weak IT/process conditions

- Manual data stitching (Excel ETL) → inconsistent time stamps, lost values, duplicated rows
- Missing data-quality metrics (completeness, timeliness, consistency) → regression looks "plausible" but is systematically biased
- "Proxy trap": optimising e.g. MAPE rather than true system cost (asymmetric costs of forecast errors)

2) Logistic regression (event classification, e.g. failure yes/no)

Mathematical definition

For binary labels $y_i \in \{0,1\}$ model $p(y = 1|x) = \sigma(\beta^\top x + b), \quad \sigma(z) = \frac{1}{1+e^{-z}}$.

Trained via maximum likelihood / minimisation of cross-entropy.

Practical example

- *Wind:* "Will a pitch fault occur within the next 7 days?" (binary event over a defined early-warning window)
- *Grid:* "Will an alarm cluster escalate into an incident?" (OT/IT events → incident label)

Data required

Event labels plus feature vectors, ideally including context (weather, operating mode, maintenance, firmware version).

Process/IT pitfalls

- Labels are rarely clean (ticket text rather than structured fault codes)
- Breaks between SCADA, CMMS and ticketing → no consistent ground truth

3) Decision trees / random forests (non-linear models, robust with mixed data)

Mathematical definition

A decision tree approximates $f(x)$ via recursive partitioning of the feature space (classification often minimises impurity such as Gini/entropy). A random forest averages many trees:

$\hat{f}(x) = \frac{1}{T}\sum_{t=1}^{T} f_t(x)$, with f_t trained on bootstrap samples and random feature subsets.

Practical example

- *Wind:* fault classification (converter / pitch / generator) from SCADA and CMS features
- *Grid:* prioritisation of alarms from protection/control systems plus telemetry

Data required

Mixed structured data (numeric + categorical) are fine; forests perform well when feature engineering is solid.

Pitfalls under weak IT/process conditions

- Without feature-store logic / versioning: "same" features computed differently across teams → non-reproducible models
- Schema changes (new SCADA columns, new alarm codes) break pipelines; without observability, the break is silent

4) Support Vector Machines (SVM) – Trennflächen / Margin-Maximierung

Mathematical definition

For labels $y_i \in \{-1, +1\}$: $\min_{w,b} \quad \frac{1}{2}|w|^2 \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1$.

With a soft margin, slack variables and a penalty parameter C are added. The kernel trick enables non-linear separation.

Practical example

- *Wind:* classification "normal vs anomalous" in feature space (e.g. spectral features)
- *Grid:* classification of operating situations (e.g. "stable vs critical")

Data required / common stumbling blocks

- SVM scales less well on very large datasets; it needs clean feature scaling
- In OT/SCADA environments, features are often not normalised or consistent → "works once, drifts later"

5) Bayessche inference (Bayesian models) - uncertainty made explicit

Mathematical definition

Treat parameters θ probabilistically and update via Bayes' rule: $p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}$.

Practical example

- *Wind forecasting*: probabilistic forecasts (confidence intervals), regime shifts (seasonality, wake)
- *Grid*: state / risk estimation under uncertain measurements

Data requirement

Not necessarily "more data", but sound likelihood/error assumptions and consistent measurement chains.

IT/process weakness

Bayesian models are sensitive to "silent data breaks": if measurement definitions change (sensor replacement, calibration), the likelihood no longer fits → the posterior becomes wrong.

6) k-means clustering (unsupervised) - identifying operating regimes

Mathematical definition

For k cluster centres μ_k: $\min_{\{\mu_j\}} \sum_{i=1}^{n} \quad \min_{1 \leq j \leq k} |x_i - \mu_j|^2$.

Practical example

- *Wind*: clusster operating regimes (e.g. "normal operation", "curtailment", "icing-like patterns")
- *Grid*: patterns of event states and load-flow situations

Data required

Normalised feature vectors; robust features (otherwise you cluster sensor artefacts).

Pitfalls without solid IT/processes

- "Garbage in, garbage out": fluctuating data quality yields seemingly plausible clusters that reflect ETL/ sensor issues
- Without lineage/ auditability, root-cause analysis becomes extremely difficult

7) PCA (Principal Component Analysis) - dimensionality reduction / feature compaction

Mathematical definition

PCA finds orthonormal directions $w_1, \ldots, w_m$ maximising variance; equivalently eigenvectors of the covariance matrix Σ for the largest eigenvalues: $\Sigma w = \lambda w$.

Practical example

- *Wind/CMS:* heavily compress spectral data; principal components as inputs to anomaly/RUL models
- *Grid:* reduce many correlated measurements (voltages/currents/phases) to a few stable factors

Data required

Continuous, scaled/centred data; PCA is sensitive to outliers.

IT/process pitfalls

If data are stale or time alignment is missing, PCA learns spurious dominant modes.

8) Neural networks / deep learning (incl. backpropagation) – universal function approximation

Mathematical definition

A feedforward network is a composition $f(x;W) = \phi_L\left(W_L\phi_{L-1}(\dots\phi_1(W_1x))\right)$; training typically uses gradient descent on a loss L: $W \leftarrow W - \eta\nabla_W \sum_i l(f(x_i;W), y_i)$.

Practical examples

- CNNs for rotor-blade images (segmentation/detection)
- RNN/LSTM/Transformer models for time series (wind power, condition monitoring, grid telemetry)
- Autoencoders (next section) for anomaly detection

Data required

Large, consistent, representative datasets. For vision: clean labels/annotations; for time series: stable sampling rates and clear masking of missing values.

Main issues under weak IT/process conditions

- Training data "move" (no versioning, no lineage) → results not reproducible
- Bandwidth/storage/compute become bottlenecks; data movement can rival training cost
- Without a disciplined MLOps pipeline, DL quickly leads to loss of trust ("worked yesterday")

9) Autoencoder / anomaly detection (unsupervised / semi-supervised)

Mathematical idea

An autoencoder learns an encoder $z = g_\phi(x)$ and decoder $\hat{x} = h_\theta(z)$, minimizing rekonstruction error:

$$\min_{\phi,\theta} \sum_i |x_i - h_\theta\left(g_\phi(x_i)\right)|^2.$$

Anomalies often appear as high reconstruction error or as latent-space outliers.

Practical example

- *Wind:* early detection of unusual CMS signatures without many failure labels
- *Grid/OT:* communication or measurement patterns deviating from normal profiles

Data required

A lot of clean "normal operation" data, with sufficient diversity (weather/load/operating modes).

IT/process traps

If "normal" data already include curtailment or misconfiguration, the model learns that as normal. If processes do not define what "normal" means (operating modes, maintenance windows, firmware versions), the approach becomes brittle.

10) Gaussian Processes (GP) & Bayesian / safe optimisation (wake steering, controller tuning)

GP core idea

A Gaussian Process defines a distribution over functions: $f(x) \sim \mathcal{GP}\left(m(x), k(x,x')\right)$. After observations, one obtains a posterior mean and variance; Bayesian optimisation selects the next point via an acquisition function balancing exploitation and exploration.

SafeOpt (core condition)

SafeOpt uses confidence bounds $u_i(x), l_i(x)$ and requires probalistic saftey, e.g. $P(f(x) \leq y_{max}) \geq P_{\text{safe}}$.

Practical example

Wake steering / controller tuning: raise AEP without violating load/vibration limits. Safe optimisation is "natural" because exploration cannot be arbitrary.

Data required

Operational/experimental data with clear attribution: control variables x (yaw offsets, controller parameters), utcome $f(x)$ (AEP/loads), plus context (wind direction, turbulence, operating mode). Without context, effects are not identifiable.

IT/process damage

- Missing availability/consistency at second/minute resolution → cannot isolate the effect of individual set-points
- Without change management: parallel parameter changes and maintenance interventions break causal attribution

11) Reinforcement learning (RL): MDP, Q-learning, DQN, PPO (only with a safety case)

Mathematical definition (MDP)

A MDP is $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. The objective is to find a policy $\pi(a|s)$ maximising expected discounted return:

$$\max_{\pi} \quad E[\sum_{t=0}^{\infty} \gamma^t r_t].$$

Q-learning (core update)

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a'} \quad Q(s',a') - Q(s,a)\right].$$

Practical example (where RL fits in theory)

Wake steering / MPC-like control decisions; dispatch/logistics as a sequential decision problem. In critical infrastructure, however, RL is typically simulation-driven (digital twin) and introduced as advisory/guardrailed automation.

Data required

Interaction data or high-quality simulation (digital twin), defined rewards, clear state definitions.

IT/process problems

- Without a twin and controlled change processes, RL becomes operationally unsafe
- “Reward hacking” is the proxy trap in its purest form: optimising a number rather than the true objective system

12) GenAI / RAG (Retrieval-Augmented Generation) – knowledge work with provenance

Technical definition (RAG)

Rather than letting an LLM answer “from its head”, context is retrieved from a controlled knowledge base:

1. Embedding functions $e(d) \in R^k$ for dokuments/chunks and $e(q)$ for the query
2. Retrieval: select top-K dokuments by similarity, e.g. cosine: $\mathrm{sim}(q,d) = \frac{e(q) \cdot e(d)}{|e(q)||e(d)|}$
3. Generation: the LLM generates an answer conditioned on the retrieved documents.

Practical example

Engineering/operations: “Show the relevant steps for converter fault class X, including lessons learned and extracts from OEM manuals” - with a strict provenance requirement.

Data required

Versioned documents (manuals, standards, as-built, tickets, reports) plus access/permissions concepts.

IT/process problems

- Without IAM/role models, audit logging and data classification, GenAI is rarely production-ready (privacy, trade secrets)
- Without document quality / single source of truth, RAG returns “sourced” answers - but from outdated or contradictory sources

Annex II - acronyms

Acronym	Name	About
AEP	Annual Energy Production	Annual energy yield of a wind farm
AI	Artificial Intelligence	Umbrella term for intelligent systems
ALARP	As Low As Reasonably Practicable	Risk reduced to a level that is as low as reasonably practicable
ANN	Artificial Neural Network	Neural network for function approximation
API	Application Programming Interface	Interface for system-to-system integration
AR	Augmented Reality	Assistance through the overlay of digital information
BESS	Battery Energy Storage System	Battery storage system for grid and market applications
BMS	Battery Management System	Cell protection, balancing, operating limits, and diagnostics
CMMS	Computerized Maintenance Management System	Maintenance planning, work orders, and asset history
CMS	Condition Monitoring System	Condition monitoring based on sensors and signal data
CNN	Convolutional Neural Network	Deep learning for image and raster data
CRPS	Continuous Ranked Probability Score	Performance metric for probabilistic forecasts
CTA	Call to Action	Prompt for action (e.g. clicking a link)
DA	Day-Ahead	Market time window for the following day
DL	Deep Learning	ML based on deep neural networks
DQN	Deep Q-Network	Q-learning with neural function approximation
EMS	Energy Management System	Dispatch and optimisation at asset or plant/farm level
ETL	Extract, Transform, Load	Data pipeline for integration and preprocessing
GenAI	Generative Artificial Intelligence	AI for the generation of text, images, and code
GNN	Graph Neural Network	Learning on networks, topologies, and graphs
GP	Gaussian Process	Bayesian surrogate model with uncertainty representation
GPU	Graphics Processing Unit	Hardware accelerator for ML and GenAI
HESS	Hybrid Energy Storage System	Combined storage system for multiple timescales
HPC	High Performance Computing	High-performance computing for training and simulation
IAM	Identity and Access Management	Roles, permissions, authentication, and access control
ID	Intraday	Market time window within the delivery day
LLM	Large Language Model	Language model for generative tasks
LLMOps	LLM Operations	Operation, monitoring, and quality assurance of LLMs
LSTM	Long Short-Term Memory	RNN variant for time series and sequences
MAPE	Mean Absolute Percentage Error	Percentage forecasting error for point estimates
MDP	Markov Decision Process	Formalisation of sequential decision problems
MICQP	Mixed-Integer Convex Quadratic Program	MIQP with convex quadratic structure
MIQP	Mixed-Integer Quadratic Program	Optimisation with integer variables and quadratic terms
MLOps	Machine Learning Operations	Operation, deployment, and monitoring of ML systems
MPC	Model Predictive Control	Predictive control with explicit constraints
MTU	Market Time Unit	Time interval used for market and balancing processes
N-1	N minus 1	Operational security despite the outage of one element
NLP	Natural Language Processing	Processing and analysis of natural language
NWP	Numerical Weather Prediction	Numerical weather forecasting based on model data
OCV	Open Circuit Voltage	Open-circuit voltage as an indicator of state of charge
OEM	Original Equipment Manufacturer	Manufacturer of original components
OT	Operational Technology	Operational systems for control, protection, and plant automation
PCA	Principal Component Analysis	Dimensionality reduction using principal components
PCC	Point of Common Coupling	Grid connection point of a plant or platform
PMU	Phasor Measurement Unit	Synchronous phasor measurement at high sampling rates
POMDP	Partially Observable Markov Decision Process	MDP with incomplete observability
PPO	Proximal Policy Optimization	Policy-gradient method in reinforcement learning
PTDF	Power Transfer Distribution Factor	Sensitivity of power flows to transfer changes
Q-learning	Q-learning	Learning value functions through Bellman updates
RAG	Retrieval-Augmented Generation	Generation based on document-grounded evidence
RL	Reinforcement Learning	Learning through feedback across sequences of decisions
RNN	Recurrent Neural Network	Neural network for sequential data
RUL	Remaining Useful Life	Remaining useful lifetime of an asset or subsystem
SCADA	Supervisory Control and Data Acquisition	Supervisory control, telemetry, and data acquisition
SLO	Service Level Objective	Target operating values for service performance (e.g. latency/error rate)
SOC	State of Charge	Battery state of charge
SOH	State of Health	Battery health condition and ageing state
SVM	Support Vector Machine	Classification by maximising the decision margin
UEBA	User and Entity Behavior Analytics	Detection of anomalies in user and system behaviour

Annex III - references

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